# Avi Gobrin

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## Education

**University of Toronto** Sep 2023 – Apr 2027
*Honours BSc, Mathematics & Its Applications Specialist, Statistics Major, Computer Science Minor* *Toronto, ON*

**Relevant Coursework:** Probability & Statistics, Multivariate Data Analysis, Nonlinear Optimization, Statistical Computation, Linear Algebra, Real Analysis, Mathematical Modelling, Graph Theory, Combinatorics, Data Structures and Algorithms

## Experience

**Nextdoor** May 2026 – Aug 2026
*Machine Learning Intern* *San Francisco, CA*

- Increased growth and weekly active users by launching Best Places for Families, an SEO ranking product covering ~300,000 US neighbourhoods, building every layer end to end: Airflow pipelines ingesting internal and Census/ACS data, a backend scoring service, and a Postgres/GraphQL serving path with a React frontend.
- Designed the scoring methodology by comparing 4 models to predict family-friendliness scores for the ~300,000 neighbourhoods with sufficient data; screened 33 metrics down to 8 final features.
- Validated ranking quality before launch through an evaluation harness: a preregistered neighbourhood adversarial test set with pairwise Bradley-Terry judgments, cross-checked by both LLM and manual review.
- Made the system self-sustaining with scheduled jobs, automated tests, and monitoring, and built Databricks tooling that lets stakeholders retune feature weights and see how any city ranks.

**Shopify** May 2025 – Aug 2025
*Software Engineering Intern* *Toronto, ON*

- Built a full-stack partner-attach system (Ruby on Rails, Calendly API) that screens each new merchant and books qualifying ones into guided onboarding with a vetted advisor. This Increased merchant activation and retention by routing over 100 merchants per day into the partner attach flow.
- Brought Shopify's entire German merchant base into VAT compliance within a month of the regulatory notice to launch. Reconciled what data Shopify already held against what merchants had to supply, integrated a difficult government tax API, and built a Rails intake form with an email flow that returns each merchant's compliance PDF.
- Tackled long-standing technical debt on the Run Team to minimize operational friction, successfully alleviating cross-team dependencies and unlocking faster feature delivery.

## Research

**University of Toronto, Department of Statistical Sciences** Aug 2026 - Present
*Undergraduate Researcher, Advisor: Prof. Ricardo Baptista (Vector Institute)* *Toronto, ON*

- Investigating a unifying rank-reduction framework linking in-context learning and fine-tuning as low-rank weight updates, connecting recent transformer interpretability work to classical multivariate statistics.

### Technical Reports

- Engagement, Performance, and Student Outcomes at the Open University (n = 23,301) [Report]
- Discovering Latent Structure in the IPIP Big Five Personality Inventory (n = 19,718) [Report]
- The Effect of Transaction Costs on Optimal Portfolio Rebalancing Frequency (Monte Carlo over 10,000 paths) [Report]
- What Makes a Wine Good? Chemical Structure and Quality in Portuguese Wines (n = 6,497) [Report]

## Projects

**Latent-Structure Modeling of Student Outcomes (OULAD)** *Research Project* [Report] 2026

- Modelled latent engagement and performance structure across 23,301 complete student-course registrations (of 25,820) via a parallel-analysis-validated 3-factor varimax model (Engagement $h^2 = 0.84$, Consistent Performance $h^2 = 0.72$) and 3-component PCA (72.9% variance); showed FA, GMM, and LDA converge on the same two latent dimensions.
- Benchmarked classifiers under 5-fold CV (logistic regression 67.0% vs. LDA 64.5% vs. 53.0% majority baseline) and showed LR lifts minority-class recall from 3.2% to 39.3%, diagnosing LDA's shared-covariance violation.

**GPT from Scratch** *Personal Project (in progress)* [Repo] 2026

- Building a character-level GPT in pure NumPy, growing one codebase from a bigram baseline into a full Transformer (multi-head causal self-attention, LayerNorm, Adam, temperature/top-k sampling); every forward and backward pass is derived on paper before being coded, with no autograd; bigram and neural-bigram models complete.

## Technical Skills

**ML & Data:** PyTorch, scikit-learn, NumPy, Pandas, SciPy, statsmodels, Matplotlib, Seaborn
**Languages, Systems & Web:** Python, R, SQL, Ruby, Java, JavaScript, Bash, HTML, CSS, Airflow, Databricks, Postgres, Linux/Unix, REST APIs, GraphQL, React, Ruby on Rails, Node.js, Flask
**Tools:** Git, GitHub, Jupyter, LaTeX, Claude Code, Cursor, Claude, LLM agents, Maven, Excel